<u>Basic prompting for reading comprehension</u>

[This transcript shows a naïve prompting approach wherein the AI provides long, often overly long responses which the user must then skim, filtering out the excess data to focus on the desired information.

To begin the process, the following paper was uploaded to ChatGT: B. Carstensen, Age–period–cohort models for the Lexis diagram, *Statist. Med.* 2007; **26**:3018–3045, DOI: 10.1002/sim.2764.

The prompt-and-response transcript appears below.]

Prompt:

In the attachment, there is a passage, "In this paper I will focus on practical aspects of analysis of data where events are derived from disease registers and population risk time from census data, typically from national statistical offices." what is "population risk time" in that passage? Please reply using quotes from the article where possible.

AI response:

In the article, **"population risk time" means the amount of *person-time at risk* in the population**, usually measured in **person-years**, and it is constructed from census data.

The article explains this directly. When discussing how population figures are used, it states:

"Population figures from census data are typically given by age and period and these have to be used to compute the *risk time (person-years)* in triangular subsets of the Lexis diagram."

So in this context:

- **"Population risk time" = "risk time (person-years)."**
- It refers to the **total accumulated time that individuals in the population spend being at risk of the event** (e.g., disease incidence or death), organized by age and calendar time.
- This person-time is derived from **census population counts**, interpolated over age and time to fit the Lexis diagram structure used in age-period-cohort models.

If you want, I can also explain how person-years are estimated from census counts or how triangular Lexis cells work.

Prompt:

You said "It refers to the total accumulated time that individuals in the population spend being at risk of the event (e.g., disease incidence or death), organized by age and calendar time." Can you give a hypothetical example?

AI response:

Sure — here is a clear, concrete hypothetical example of **population risk time** (i.e., *person-years at risk*) in the sense used in the article.

Hypothetical example: lung cancer incidence in 40–44-year-old men in 1990

Setup

Suppose we want to compute the lung-cancer **incidence rate** for:

- **Age group:** 40–44
- **Calendar year:** 1990
- **Sex:** Men

The numerator of the rate is the number of new cases in that age–period cell.

The **denominator** is the **population risk time**, meaning the *total amount of time that all men aged 40–44 spend alive (and therefore at risk of being diagnosed) during 1990.*

Hypothetical data

Assume national census data give:

- Population of 40–44-year-old men on **Jan 1, 1990**: **100,000**
- Population of 40–44-year-old men on **Jan 1, 1991**: **98,000**

So during the year, some men died, some aged out, some moved in/out.

Step 1 — Estimate person-years (risk time)

A simple (but slightly crude) estimate is the average of the beginning and end populations:

Risk time≈100,000+98,0002=99,000 person-years\text{Risk time} \approx \frac{100{,}000 + 98{,}000}{2} = 99{,}000 \text{ person-years}Risk time≈2100,000+98,000=99,000 person-years

Meaning:

Across the entire country, men aged 40–44 collectively spent about **99,000 years of life** during the single calendar year 1990.

More precise Lexis-triangle methods (as used in the article) adjust for age progression and mortality within the year, but the conceptual meaning is the same.

Step 2 — Compute a rate

Suppose **120 new lung cancer cases** were diagnosed in this group in 1990.

Incidence rate=12099,000≈0.00121 per person-year\text{Incidence rate} = \frac{120}{99{,}000} \approx 0.00121 \text{ per person-year}Incidence rate=99,000120 ≈0.00121 per person-year

Often expressed per 100,000 person-years:

121 cases per 100,000 person-years121 \text{ cases per 100,000 person-years}121 cases per 100,000 person-years

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What this example illustrates

- "Population risk time" = summing up **how long** all individuals in a population spend in a given age-period cell.
- It is not counting people; it is counting **time lived**.
- It is used as the denominator of rates because the *opportunity* for an event depends on how much time people spend at risk.

If you want, I can also give an example involving Lexis triangles specifically (e.g., splitting by birth cohort within an age–period cell), which is what the paper focuses on.

Prompt:

Does the paper use the term "Lexis triangle" much?

AI response:

Yes — the paper uses the term **"Lexis triangle(s)" quite a lot**, though sometimes it appears indirectly through descriptions like "triangles A and B" or "triangular subsets" of the Lexis diagram.

Here are the relevant findings from the file search:

Direct mention of "Lexis triangles"

The paper explicitly uses the phrase **"Lexis triangles"** at least **twice**, for example:

- In **Appendix A**, the heading itself is:

  > **"APPENDIX A: PERSON-YEARS IN LEXIS TRIANGLES"**

- Throughout Appendix A, triangles A and B are repeatedly referred to as parts of the **Lexis diagram**, which together form *Lexis triangles* used to compute person-years.

Related phrases used frequently

The paper also uses closely related terminology many times, including:

- **"triangular subsets of the Lexis diagram"**
  Example:

  > "Such figures … can be used to compute the risk time (person-years) in **triangular subsets of the Lexis diagram**."

- References to **"triangle A" and "triangle B"**
  These occur many times where person-time is calculated separately for each triangle.

- **Figure 3**, which visually depicts the two Lexis triangles (A and B).

Frequency summary

- The exact phrase **“Lexis triangles”** appears **a small number of times**, mostly in Appendix A.
- But the concept (triangular Lexis cells) is **used heavily** throughout the person-years calculations, where the risk time is computed **separately for each Lexis triangle**.

So although the exact words don’t appear constantly, the **Lexis triangle concept is central**, and triangular Lexis subdivisions are used repeatedly in formulas and figures.

If you want, I can list every exact page/line where “Lexis triangle(s)” appears, or explain why the Lexis triangle structure is essential for precise person-time estimation.

Prompt:

What does the author mean by the term "factor model"? use quotes from the paper where possible, and be quite brief.

AI response:

The author uses **“factor model”** to mean the *classical APC model in which age, period, and cohort are treated as categorical factors*, with **one parameter for each category level**.

He describes it this way:

- The “classical” APC model is

> **“formulated as a model for this table where the effect of age, period and cohort are modelled as *factors*, i.e. with one parameter per level in the tabulation.”**

He contrasts this with smoother models that treat age, period, and cohort as continuous variables.

So “factor model” here simply means:
An APC model treating age, period, and cohort as categorical variables with separate parameters for each category (e.g., 5-year age groups, 5-year periods, etc.).

Prompt:

If age is a continuous variable, does he mean that it only has one or a very few parameters? Like if a linear curve, then the only parameters would be slope and intercept?

AI response:

Short answer: Not necessarily only one or two parameters.
When Carstensen says age, period, and cohort should be treated as **continuous variables**, he does *not* mean that they must be modeled with a simple linear function (slope + intercept). He

means that you should **use smooth functions** (splines, polynomials, etc.) rather than a separate parameter for every discrete age group.

Here's how this fits the paper:

1. What he is contrasting with

He criticizes the *factor model*, where:

age, period and cohort are "**modelled as factors, i.e. with one parameter per level**"

Example: 20 age groups → 20 parameters.

2. What he means by "continuous"

Elsewhere he says one should treat age, period, and cohort as continuous and use smoothness:

"age, period and cohort should be regarded as **continuous variables**"

In practice, treating them as continuous means:

- You use **splines** (e.g., cubic splines, natural splines)
- Or **polynomials**
- Or other smooth functions
  rather than individual dummy variables.

These continuous representations typically involve **several parameters**, but *far fewer* than one-parameter-per-age-group.

Prompt:

Does he mean a simple linear model?

AI response:

No.
He does *not* imply that age should be modeled only with a straight line (slope + intercept). Linear age effects are almost never realistic in epidemiology. The paper itself uses spline-based smooth APC models.

So:

- Linear model → 2 parameters
- Spline model → usually 4–10 parameters
- Factor model → possibly 20–40 parameters

Carstensen wants models in the **middle**: smooth but flexible.

Bottom line

When he says age/period/cohort should be continuous, he means:

➡️ **Use smooth functional forms (splines), not simple linear functions, and not a huge number of factor levels.**

If you want, I can summarize how the Epi package (which he wrote) parameterizes continuous age/period/cohort effects.

Prompt:

Very briefly, for dummies, what is a spline?

AI response:

A **spline** is just a **smooth, bendable curve** made by joining simple curve pieces (usually polynomials) together so that it looks smooth.

Think of it like:

A flexible ruler □ you can bend it gently to follow the shape of the data, without making sharp corners or wild zigzags.

It's more flexible than a straight line, but not as wiggly as fitting every point individually.

Prompt:

Each curve piece has a different functional form?

AI response:

Not a different form — each piece is usually the **same type of polynomial** (e.g., cubic), but the **coefficients change** from piece to piece.

So:

- All pieces use the same formula type (e.g., a cubic polynomial).
- They are joined at "knots."
- The joins are forced to be smooth (no sharp corners).

So a spline is **many small cubic curves stitched together smoothly**, not many different kinds of formulas.

Prompt:

Here is a sentence from the article: "Rates versus age, observations within each period connected, i.e. cross-sectional age-specific rates." Does this mean 3 dimensions of data: rate, age, and period?

AI response:

Yes.
That sentence refers to **three dimensions of data**:

- **Rate** (the outcome being plotted)
- **Age** (x-axis of the plot)
- **Period** (which curve you are looking at)

Here is what the author means:

"Rates versus age, observations within each period connected, i.e. cross-sectional age-specific rates."

This describes a plot where:

- On the **x-axis**: **age**
- On the **y-axis**: **rate** (e.g., incidence rate)
- **Separate curves**: one for each **period** (calendar year or calendar interval)

So the data are indeed three-dimensional (rate × age × period), but the graph shows only a **2D slice** at a time (rate vs. age), with **multiple lines**, each line representing a different period.

Prompt:

Are you sure?

AI response:

Yes — I am sure, and here is the clearest way to see it.

What the sentence describes

"Rates versus age, observations within each period connected, i.e. cross-sectional age-specific rates."

This describes a *plot*, not the underlying raw data structure.

Plot structure

- **x-axis: Age**
- **y-axis: Rate**
- **One line per period:** all age-specific rates *for that period* connected

So the **data feeding the plot** must have three variables:

1. **Age**
2. **Period**
3. **Rate** (cases / person-years)

The graph itself is 2-dimensional (rate vs. age), but you draw **multiple curves**, one for **each period**, which requires that the underlying data have all **three** dimensions.

Why this is correct

Carstensen uses “cross-sectional age-specific rates” to refer to rates observed **across ages at a single calendar period**. To draw those, you need:

- Age (horizontal)
- Rate (vertical)
- Period (which line you’re drawing)

So yes — the sentence describes a visualization based on **three-dimensional data**:
(age, period, rate).

Prompt:

So, hold period constant and see how rate changes with age; then plot a similar curve but for a different period?

AI response:

Exactly — that’s precisely what the sentence means.

How it works

1. **Pick one period** (say 1980).
   - Look at the **age-specific rates** for that period.
   - Plot **rate vs. age**.
   - Connect the points → this is one *cross-sectional* curve.
2. **Pick another period** (say 1990).
   - Do the same: plot **rate vs. age** for 1990.
   - Connect those points → a second curve.
3. **Repeat for other periods**.
   - Each period produces its own “cross-sectional age curve.”

So the figure has:

- **One x-axis:** age
- **One y-axis:** rate
- **Multiple curves:** one for each period

This shows how the *shape of age-specific rates* changes across different calendar periods.

Prompt:

So by “cross-sectional age curve” is meant, across all age groups?

AI response:

Yes.
A “cross-sectional age curve” means **a curve showing the rate across all age groups, but for one specific period.**

In Carstensen’s usage:

- **Cross-sectional** = looking at *different ages at the same time* (same period).
- **Age curve** = a plot of **rate vs. age**.
- So a **cross-sectional age curve** = rate across all ages *within a single period*.

Then you can draw **another** cross-sectional age curve for a different period, and compare them.

Prompt:

Consider the next sentence, "Rates versus age, observations within each birth cohort connected, i.e. longitudinal age- specific rates." Isn't it a problem to get these curves because birth cohorts more recent than about a hundred years ago don't have a complete cross section because some age groups are still alive?

AI response:

Great question — and **yes**, that issue is real, but it's *expected* and *not a problem* for the kind of plot Carstensen is describing.

Let me explain briefly and clearly.

✔ What the sentence means

“Rates versus age, observations within each birth cohort connected, i.e. longitudinal age-specific rates.”

This means:

- Pick **one birth cohort** (e.g., people born in 1950).
- For that cohort, plot **rate vs. age**.
- Each point corresponds to the rate the cohort experienced **at that age**, which happens in different calendar periods.
- Connect the points → this yields a *longitudinal* age curve.

✔ Your concern

You are noticing that:

- A cohort born in (say) 1980 has only reached age ~45 by 2025.
- So we **cannot** draw a full age curve (e.g., ages 0–100) for that cohort yet.

Correct! That is absolutely true.

✔ Why this is *not* a problem

Carstensen expects this. The longitudinal curves for birth cohorts are **not required to be complete**. They simply:

- **start** when data begins (for older ages, maybe 1950 or so),
- **end** at whatever the cohort's current age is,
- and exist only where data exist.

For example:

Cohort	Maximum age observed	Curve completeness
Born 1930	~95	Nearly full curve
Born 1950	~75	Partial curve
Born 1980	~45	Shorter curve
Born 2000	~25	Very short curve

This is **normal** and seen in all APC plots.

Carstensen is not claiming every cohort yields a full 0–100 age trajectory. Instead:

- Older cohorts have long curves.
- Younger cohorts have short curves.
- All are still useful because they show how rates change *within* each cohort over the ages for which we have data.

✔ Why incomplete curves are still meaningful

Because a “longitudinal” age curve simply means:

Rates observed for the same cohort at different ages.

Even if the cohort has only reached age 30, the part of the curve up to age 30 is valid.

This helps identify cohort-specific patterns, such as:

- cohorts with unusually high rates at young ages,
- generational effects,
- changes in exposure history.

✔ Bottom line

Yes, younger cohorts give incomplete curves — but that is expected and not a flaw.

Longitudinal (cohort-based) age curves are **only plotted over the ages for which data exist**. Older cohorts have longer curves; younger cohorts have shorter ones.

If you like, I can illustrate this with a small hypothetical example or describe how these curves look when plotted together.

Prompt:

So, longitudinal means "over time"?

AI response:

Yes — exactly.

In this context, **longitudinal** means:

Following the *same birth cohort* over time, as they age.

So:

- **Longitudinal age-specific rates** = rates observed at
  age 20 → then age 30 → then age 40 → etc.
  for the **same cohort** (e.g., people born in 1950).

This is "over time" in the sense that:

- The cohort ages as calendar time passes (e.g., 1950 + 20 years = 1970, 1950 + 30 years = 1980).

So yes — longitudinal = **tracking the same group across different times/ages**, rather than comparing different groups at one time (which would be cross-sectional).

So by cross-sectional is meant each data point in the curve refers to a different group of people (i.e. across groups); but longitudinal means you keep observing the same group of people?

Yes — that's exactly right.
And that distinction is *precisely* what Carstensen is invoking.

Here is the clearest way to put it:

✔ Cross-sectional

Each point comes from a *different* group of people.

Example:

- Rate at age 30 in 1990 → people born in 1960
- Rate at age 40 in 1990 → people born in 1950
- Rate at age 50 in 1990 → people born in 1940

These are **different cohorts**.
So when you plot *rate vs. age within a single period*, each age point refers to a different birth cohort.

This is a **cross-sectional age curve**, because you are taking a *cross-section of the population at one moment in time*.

✔ Longitudinal

Each point comes from the *same* group of people — one birth cohort — observed at different ages/times.

Example (cohort born in 1950):

- Rate at age 20 (in 1970)
- Rate at age 30 (in 1980)
- Rate at age 40 (in 1990)
- Rate at age 50 (in 2000)

All points refer to the **same birth cohort**, just at different ages and calendar periods.

This is a **longitudinal age curve**, because you follow the *same generation* through time.

✔ Summary (your words — fully correct)

Cross-sectional: each data point comes from a different group.
Longitudinal: you keep observing the same group.

Exactly right.

Prompt:

For the next sentence, "Rates versus period, observations within each age-class connected." Each curve is for a different age class?

AI response:

Yes — that's exactly what it means.

"Rates versus period, observations within each age-class connected."

This describes a plot where:

- **x-axis:** period (calendar time)
- **y-axis:** rate
- **Each curve:** one **age class** (e.g., ages 30–34, 35–39, 40–44)

So you:

1. Pick an age class (say 40–44).
2. Look at its rate in each period (e.g., 1980, 1985, 1990).
3. Connect those points → **one curve**.

Then repeat for another age class, producing another curve.

So yes:

Each age class gets its own line, showing how the rate for that age group **changes over calendar time**.

Prompt:

So each curve is cross-sectional?

AI response:

No — in this case each curve is *not* cross-sectional.
It is actually **neither fully cross-sectional nor longitudinal in the strict cohort sense**, but here is the clean way to think about it:

✔ What each curve represents

For:

"Rates versus period, observations within each age-class connected."

You choose **one age class** (say 40–44) and plot:

- Rate at age 40–44 in 1980
- Rate at age 40–44 in 1985
- Rate at age 40–44 in 1990
- Rate at age 40–44 in 1995
- … and so on

Then connect them.

✔ Are these points from the same people?

No.
The age class stays fixed, but the cohort changes:

- Age 40–44 in 1980 → cohort born 1936–1940
- Age 40–44 in 1990 → cohort born 1946–1950
- Age 40–44 in 2000 → cohort born 1956–1960

Each point is a **different cohort**, just like in the cross-sectional age curves.

✔ But is it "cross-sectional"?

Not in the usual sense of a "cross-sectional age curve," because:

- In a cross-sectional age curve you hold **period fixed** and change **age**.
- Here you do the *opposite*: hold **age fixed** and change **period**.

Carstensen does not call these "cross-sectional," because the word is reserved for the age-vs-age comparisons at a single time.

He simply calls these:

Period trends within age classes.

✔ Best description

Each curve is:

A time series of rates for a single age group.

Or more formally:

- **Same age class**
- **Different periods**
- **Different underlying cohorts**

✔ Summary

Curve type	What varies	What is fixed	Cohort consistency?	Usual name
Cross-sectional age curve	Age	Period	No	"Cross-sectional"

Curve type	What varies	What is fixed	Cohort consistency?	Usual name
Longitudinal cohort curve	Age	Cohort	Yes	“Longitudinal”
Period trend within age class	Period	Age	No	*Not* called cross-sectional

So:

No, these period-trend curves are not cross-sectional.
Each curve follows *the same age class* over time, but *different cohorts*.

Prompt:

For "Rates versus cohort, observations within each age-class connected." Each data point in a curve is for a different cohort. But it is not cross-sectional because "cross-sectional" means period is constant? And by holding age constant and changing cohort, you necessarily change period too? Because age = period - cohort?

AI response:

Exactly — you’ve got the logic **perfectly right**, and your reasoning matches the structure of the Lexis diagram.

Let me confirm each part clearly:

✔ 1. Each data point is for a different cohort

Yes.
For a fixed age class (say 40–44):

- Cohort = period − age
- So as you vary **cohort**, you must also vary **period**.

Example (age 40–44):

Cohort	Period (approx.)	Age
1930	1970–1974	40–44
1940	1980–1984	40–44
1950	1990–1994	40–44

So each point in the curve comes from a **different cohort**.

✔ 2. It is *not* called cross-sectional

Correct.

Why not?

In APC terminology:

- **Cross-sectional** means: **period fixed, age varies**
  (Different cohorts at one moment in time.)

But here:

- Age is fixed
- Cohort varies
- Therefore period must also vary (because *age* = *period* − *cohort*)

Thus, **period is *not* held constant**, so this is *not* a cross-sectional curve.

✔ 3. Why holding age constant implies period must change

Because of the identity:

age=period−cohort.\text{age} = \text{period} - \text{cohort}.age=period−cohort.

So if you fix age and change cohort:

- A change in cohort forces a change in period.

Therefore, it is impossible to hold both age and period constant while varying cohort.

✔ So what *is* this curve, conceptually?

It is:

A cohort plot at a fixed age.

Or:

How rates differ across cohorts for the same age.

This is often interpreted as showing:

- **Generational differences**
- Differences in early-life exposure histories

- Cohort-specific risk patterns

✔ Summary (your words, validated)

Each data point is a different cohort.
It is not cross-sectional because cross-sectional means period is constant.
And if age is constant and cohort changes, period must also change (since age = period − cohort).

All of that is correct.

Prompt:

Consider the sentence "These four plots are initial explorations of whether rates are proportional between periods or cohorts." Using quotes from the paper where possible, what is meant by "proportional"?

AI response:

Here is what **"proportional"** means in that sentence, using the author's own words where available and explaining the idea briefly and accurately.

✔ What "proportional" means in the paper

Carstensen uses **proportionality** to mean that **entire curves differ only by a constant multiplier**—that is, the **shapes are the same**, but the **levels are shifted up or down** by a constant factor.

This is visible when he defines *rate ratios* between periods and cohorts. In Section 2 he says:

"Rate ratios… express *proportional differences* between rates."

A *rate ratio* being constant across ages or periods is exactly what he calls **proportionality**.

✔ In plainer terms

Two rate curves (for example, for two periods or two cohorts) are **proportional** if:

- One curve looks like the other
- Just moved up or down by a constant ratio
- No change in shape

Example:

- Period A has rate curve: r(a)r(a)r(a)
- Period B has rate curve: 1.3 r(a)1.3 \, r(a)1.3r(a)

These two are **proportional**.
Every point differs by **the same factor (1.3)**.

✔ Why the four plots check this

Earlier, he describes four basic plots (cross-sectional, longitudinal, period-specific, cohort-specific). Then he writes:

"These four plots are initial explorations of whether rates are *proportional* between periods or cohorts."

Meaning:

- In the cross-sectional and longitudinal age curves,
  Are the curves for different periods parallel in shape?
- In the period-specific and cohort-specific plots,
  Are the curves for different age classes or cohorts parallel?

If they *are* parallel, you can model them with **multiplicative effects**—i.e., APC models with smooth additive components on the log scale.

✔ Summary

"Proportional" means that rate curves for different periods or cohorts have the same shape and differ only by a constant multiplicative factor (i.e., a constant rate ratio).

This is what the author is visually checking in those initial plots.

Prompt:

The paper says "These plots require a reasonably coarse tabulation to be informative. But the tabulation used for a first simple overview of data need not be the basis for the entire analysis." Does this mean that the bins have to be big enough to have enough samples in them to smooth out the noise?

AI response:

Yes — that is exactly what he means.

Here is the sentence with context:

"These plots require a reasonably coarse tabulation to be informative. But the tabulation used for a first simple overview of data need not be the basis for the entire analysis."

✔ What “reasonably coarse tabulation” means

It means **using wider bins** (e.g., 5-year age groups, 5-year periods) so that:

- you have **enough person-years** in each cell,
- the **rates are stable**, and
- the plots are **not dominated by random noise**.

In other words:

Yes — bins must be big enough that each age–period or age–cohort cell has enough data to produce a smooth, interpretable curve.

Small bins (e.g., 1-year age × 1-year period) often produce noisy, jagged plots, especially for rare diseases.

✔ Why he emphasizes this

He is saying:

- For **initial exploratory plots**, use coarse bins to see broad patterns without noise.
- But later, for the **full APC model**, you can switch to a finer tabulation or even work without tabulation (using continuous age/period/cohort functions).

✔ Short version

Yes: coarse = enough data in each bin to smooth out noise.

[… etc.]